

Answer **all** questions in the answer booklet provided. Please start each question on a new page. Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

1. [Maximum mark: 23]

**This question asks you to use polynomial functions to model some situations in probability.**

Two unbiased tetrahedral (four-sided) dice with faces labelled 1, 2, 3 and 4 are thrown and the scores recorded.

The random variable  $M$  denotes the maximum of these two scores.

The probability distribution of  $M$  is given in the following table.

| $m$        | 1              | 2              | 3              | 4              |
|------------|----------------|----------------|----------------|----------------|
| $P(M = m)$ | $\frac{1}{16}$ | $\frac{3}{16}$ | $\frac{5}{16}$ | $\frac{7}{16}$ |

(a) Find  $E(M)$ . [2]

An alternative way to represent the probability distribution of  $M$  is to use a polynomial

function,  $G$ , where  $G(t) = \sum_{m=1}^4 P(M = m)t^m$ .

Hence, for the distribution of  $M$ ,  $G(t) = \frac{1}{16}t + \frac{3}{16}t^2 + \frac{5}{16}t^3 + \frac{7}{16}t^4$ .

(b) Find  $G(1)$ . [1]

(c) (i) Find  $G'(t)$ . [2]

(ii) Hence, show that  $G'(1) = E(M)$ . [3]

**(This question continues on the following page)**

**(Question 1 continued)**

A bag contains two red balls and three yellow balls.

Two balls are selected at random without replacement from the bag.

The random variable  $X$  denotes the total number of red balls selected.

The probability distribution of  $X$  can be represented by the polynomial function,  $G_X$ , where

$$G_X(t) = \sum_{x=0}^2 P(X=x)t^x.$$

- (d) Show that  $G_X(t) = \frac{3}{10} + \frac{3}{5}t + \frac{1}{10}t^2$ , making it clear how the coefficients of  $G_X(t)$  have been determined. [5]

An unbiased coin and a biased coin are tossed.

The probability of obtaining a tail on the biased coin is  $p$ .

The random variable  $Y$  denotes the total number of tails obtained from tossing both coins.

The probability distribution of  $Y$  can be represented by the polynomial function,  $G_Y$ , where

$$G_Y(t) = \sum_{y=0}^2 P(Y=y)t^y.$$

- (e) Given that the coefficient of  $t^2$  in  $G_Y(t)$  is  $\frac{1}{3}$ , find

(i) the value of  $p$ ; [2]

(ii) an expression for  $G_Y(t)$ . [4]

The random variable  $Z$  denotes the sum of the total number of red balls selected,  $X$ , and the total number of tails obtained from tossing both coins,  $Y$ .

The probability distribution of  $Z$  can be represented by the function,  $G_Z$ , where

$$G_Z(t) = G_X(t)G_Y(t).$$

- (f) For random variable  $Z$ , it can be shown that  $G'_Z(1) = E(Z)$ .

Use this result to find  $E(Z)$ . [4]